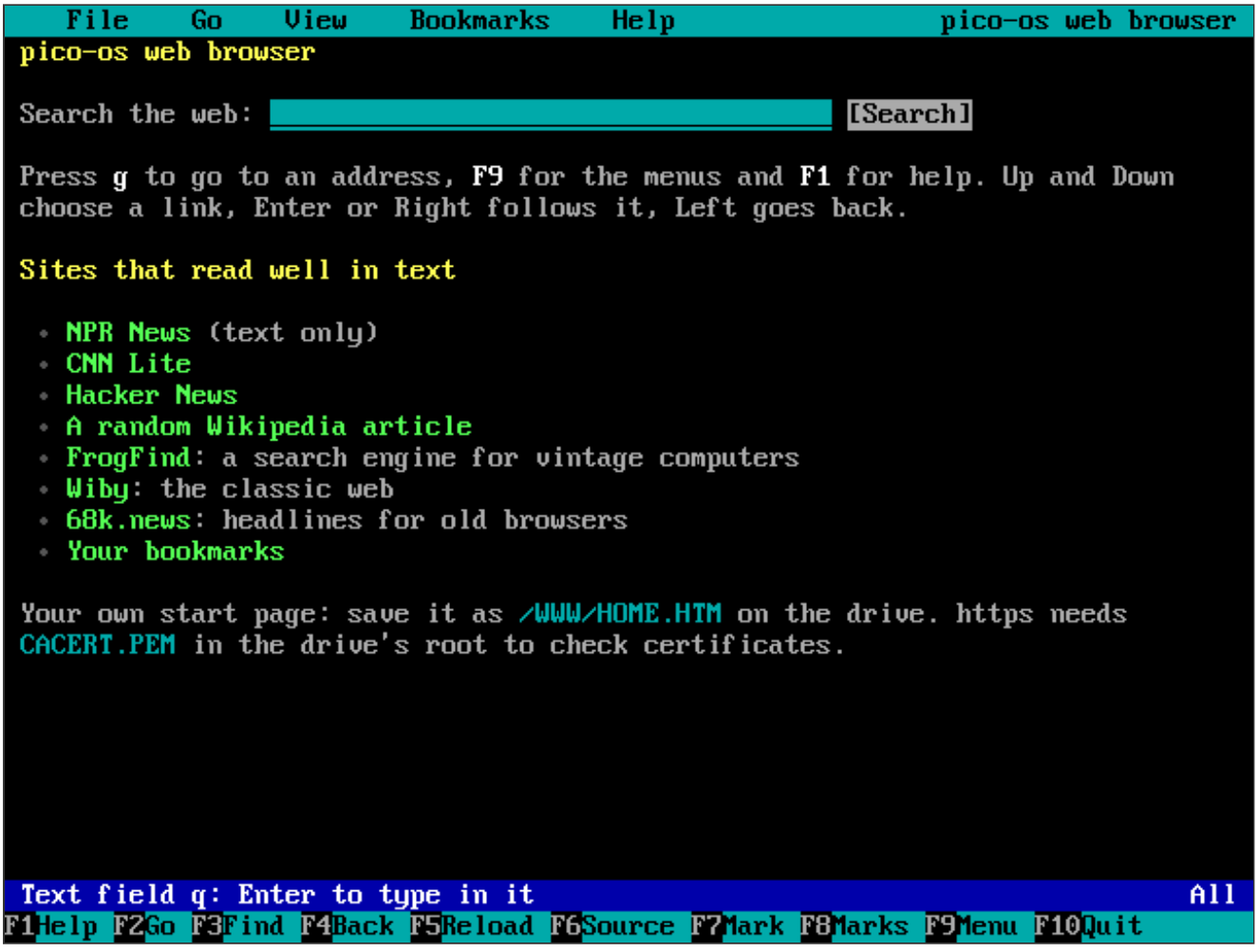
Pico 2 mini OS Web Browser

Pico 2 mini OS · October 29, 2026

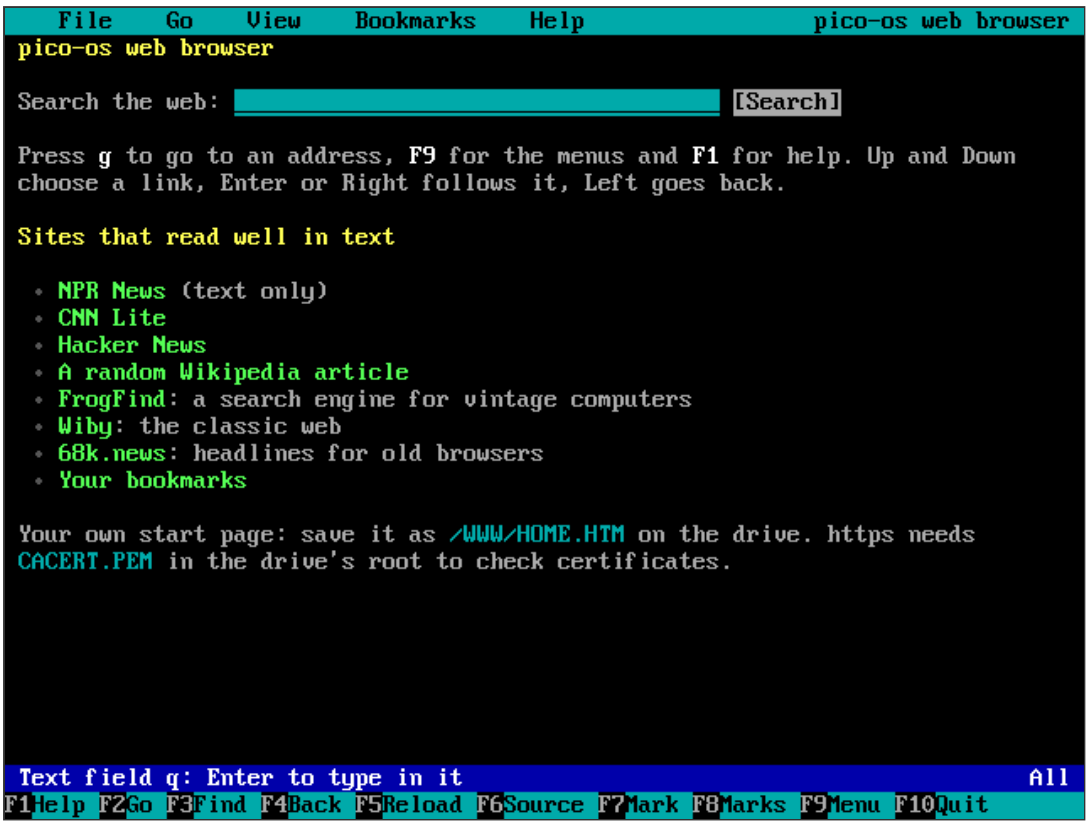


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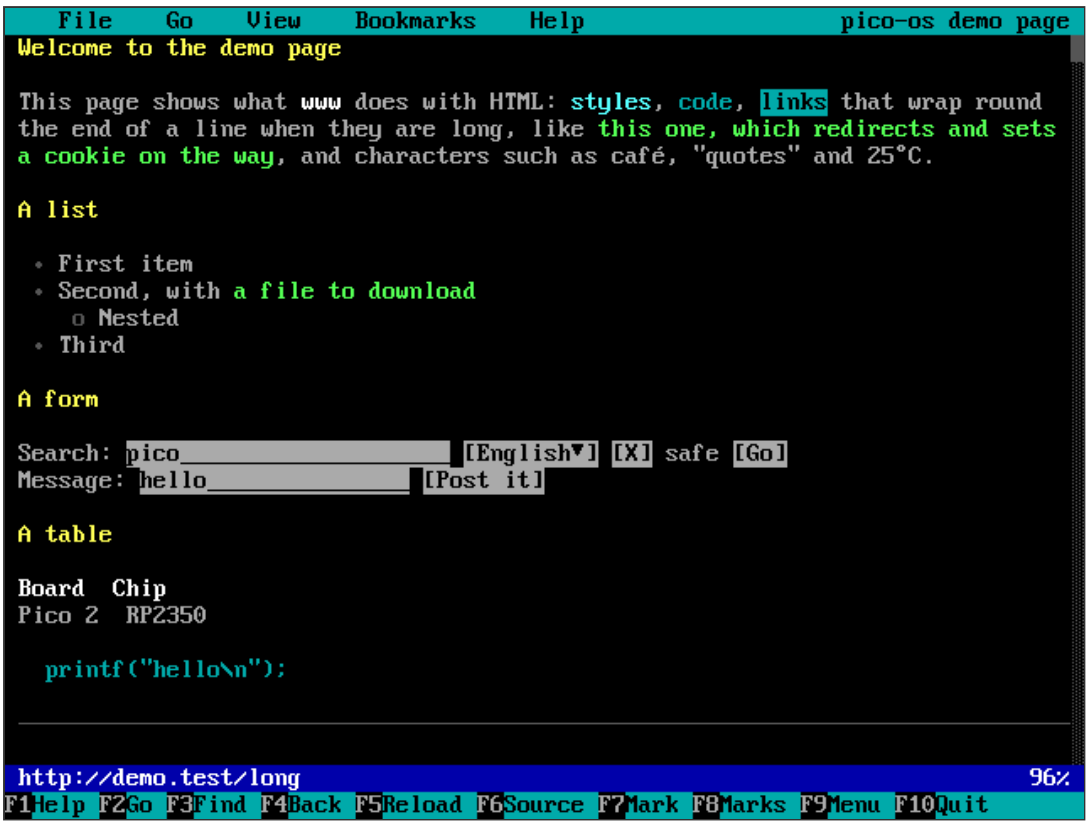
`www` (or `lynx`) is a text web browser in the manner of lynx: it loads pages over http and https (the same network and TLS code as `wget`) and shows them as text on the 80x30 screen, with their links and simple forms. Pictures are shown one at a time on the whole screen when chosen; there are no scripts or styles; most news, reference and search pages read well. It needs WiFi (`wifi connect`), and for https the certificate file `CACERT.PEM` in the drive's root (as for `wget`).

Starting



- `www` opens the start page: `/WWW/HOME.HTM` on the drive if there is one, else a built-in page with a search box and some sites that work well in text.
- `www text.npr.org` (or `www https://text.npr.org/`) opens that address. Without `http://` or `https://`, https is used.
- `www /DOCS/PAGE.HTM` opens a page on the drive (also `file:///...`).
- `www raspberry pi pico` searches the web for those words (DuckDuckGo's HTML page).

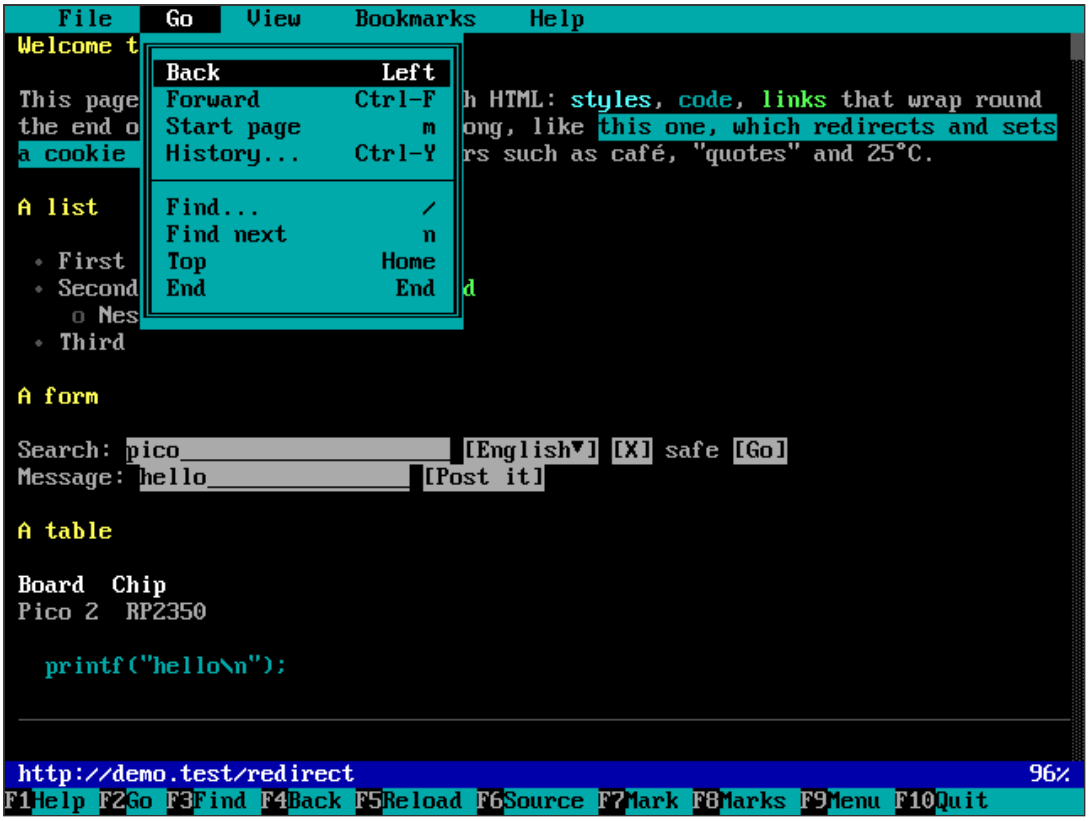
The screen



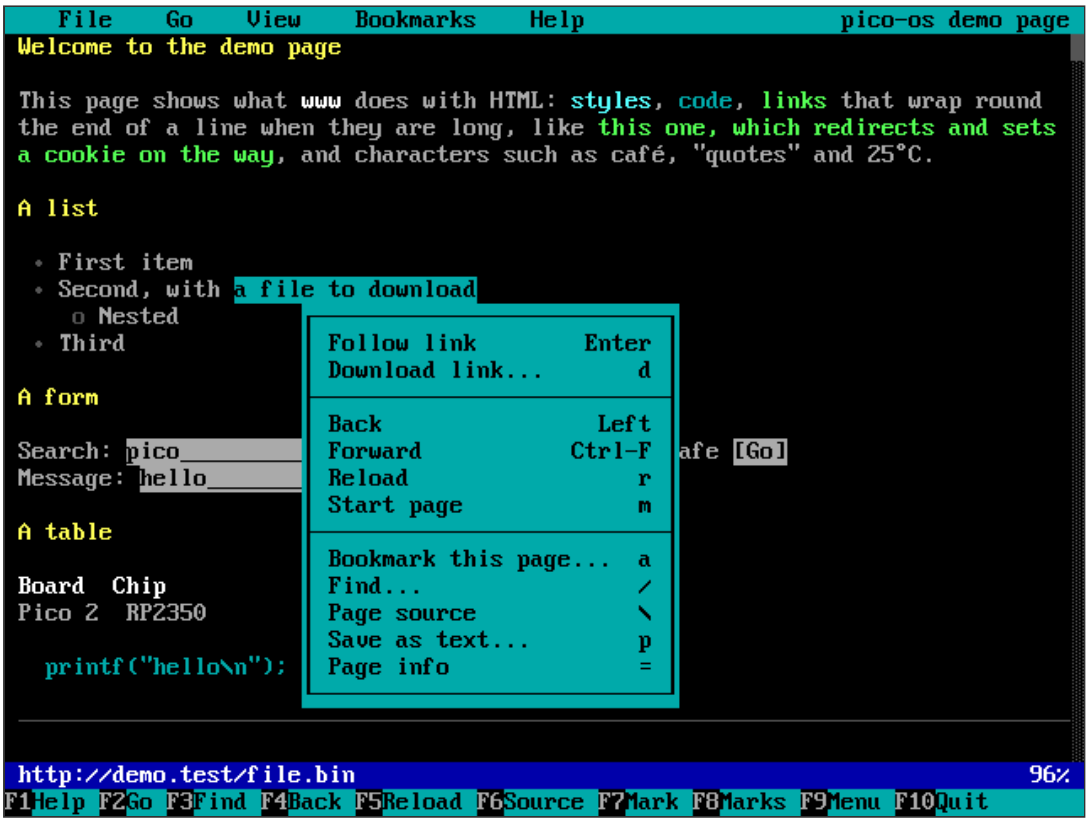
- **The menu bar** at the top: File, Go, View, Bookmarks and Help, as in mc and the tracker. F9 (or Alt and a title's first letter, or a click) opens it. The page's title is on the right.
- **The page**, 78 characters wide. Headings are yellow, bold text white, italics light cyan, code cyan, a picture's alt text magenta in `[brackets]`. **Links are green**; the link chosen is black on cyan. Form fields are grey, with `____` for the room in a text field, `[X]` for a checkbox, `(*)` for a radio button and `[choice▼]` for a list.
- **The scroll bar** on the right (when the page is longer than the screen): click it to jump there.
- **The status line** says where the chosen link goes (or what the form field is), and how far down the page is shown. While a page loads it shows the progress: looking up the name, connecting, the TLS handshake, the kilobytes so far. Esc stops loading and shows what arrived.
- **The key bar** at the bottom: click a key to press it.

Keys

Key	
Up, Down, Tab	The previous or next link. When the next one isn't on the screen, the page scrolls half a screen.
Enter, Right	Follow the link, or work the form field (below).
Left, Backspace, F4	Back. Ctrl-F goes forward again.
PgDn, Space / PgUp, b	A screen down / up.
Home / End	The top / the end of the page.
g	Go to an address (or type words to search for them).
G, F2	Edit this page's address and go there.
o	Open a file on the drive.
/, F3	Find text on the page (any case); **n** finds it again (from the top after the end).
\, F6	The page's source (the HTML as it came), and back.
r, F5	Load the page again.
m	The start page.
a, F7 / v, F8	Bookmark this page / show the bookmarks.
d	Download the chosen link's file to the drive.
s / p	Save the page's source / save it as text (as shown).
=	About the page and the chosen link.
i	Show the chosen picture, or the chosen link's picture, on the whole screen.
Ctrl-Y	The history: pick a page to go back (or forward) to.
h, F1	Help (the keys).
q, F10	Quit.



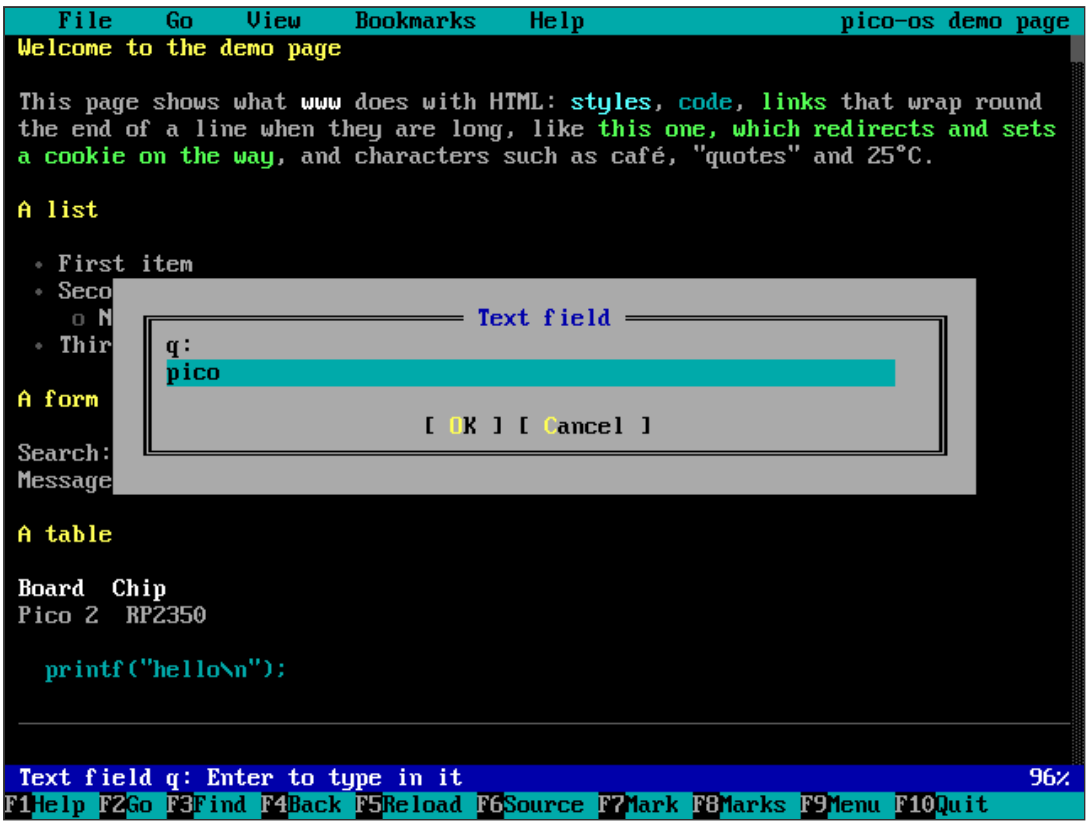
The mouse: click a link to follow it; the wheel scrolls; click the menu bar, the key bar or the scroll bar.



The right button opens a menu where you click: Back, Forward, Reload, Start page, Bookmark this page, Find, Page source, Save as text and Page info, and over a link also Follow link and Download link (the link is chosen), and View image when it is a picture or has

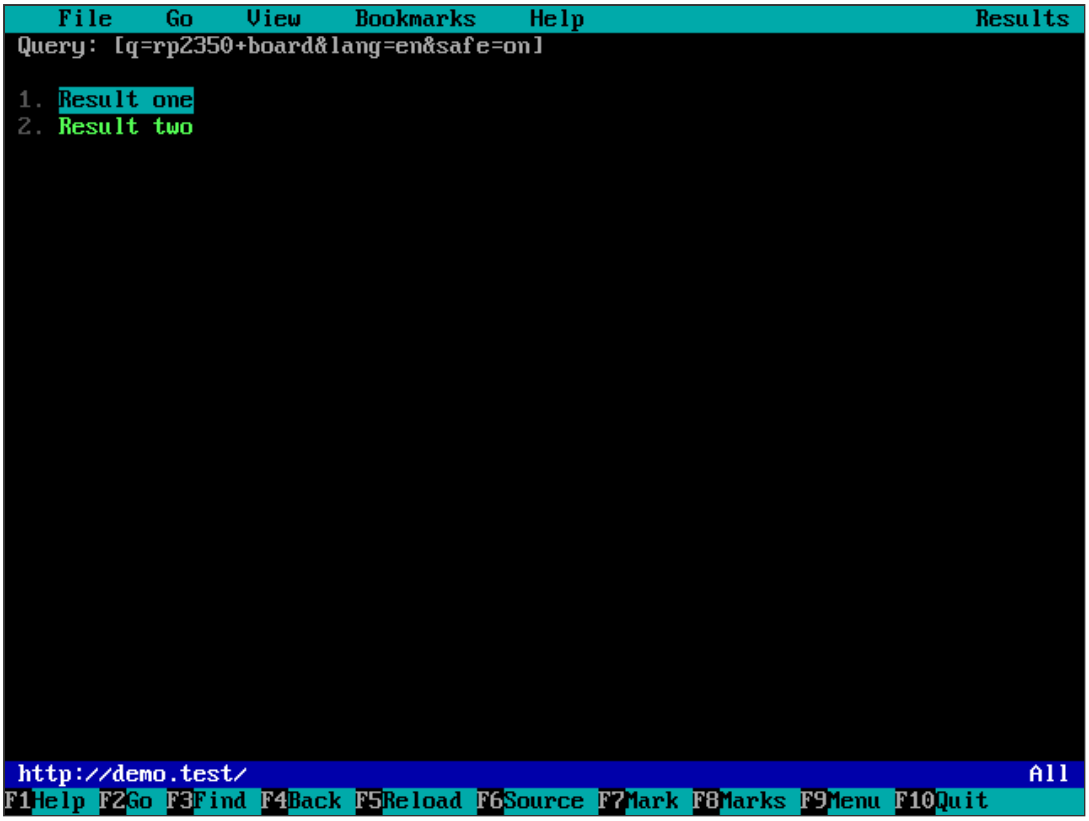
one. Click an item, or hold the right button, move to an item and let go; Up, Down and Enter work too; Esc or a click outside closes it.

Forms



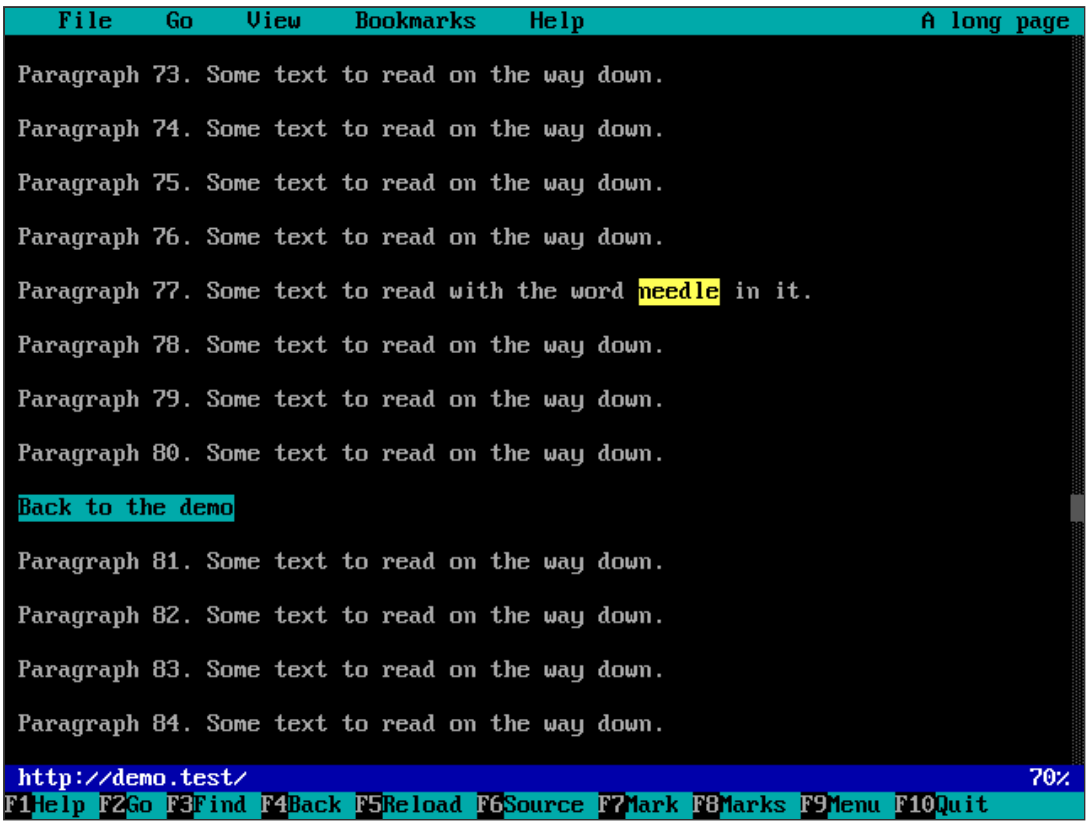
Search boxes and simple forms work. Choose a field with Up and Down, then Enter:

- **A text field** (or a password, or a text area) opens a box to type in (Ctrl-U clears it). In a form with only one text field, such as a search box, Enter in the box sends the form at once.
- **A checkbox** is ticked or cleared; **a radio button** is chosen (the others of its group are cleared).
- **A list** opens the choices to pick from.
- **A button** (`[Search]`, `[Go]`) sends the form, by GET or POST as the page asks.



Cookies a site sets are kept (in PSRAM) until the browser closes, so logins and sites that need a cookie work while it is open.

Finding text



`/` asks for the text; the page scrolls to it and it is shown in yellow (and a link on that line is chosen). `n` finds the next one.

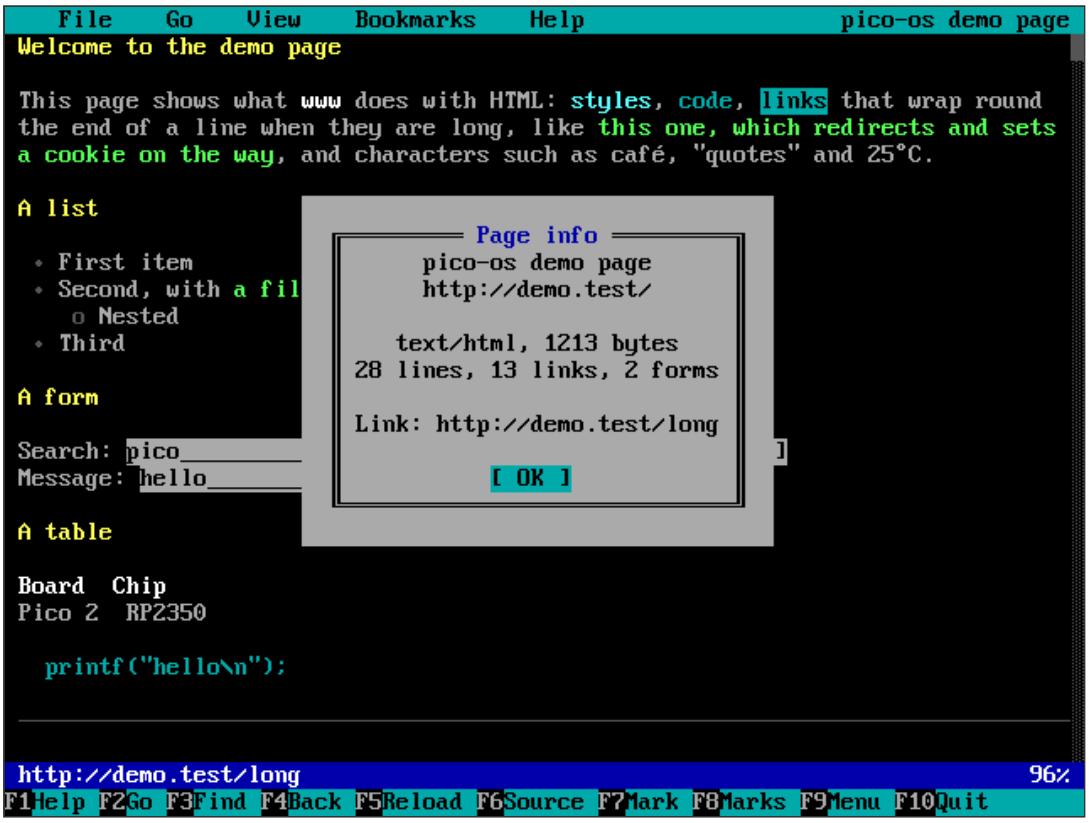
Pictures

Pages show a picture's alt text in `[brackets]`. A picture with alt text that isn't in a link is a link of its own (magenta): choose it and press Enter (or `i`) to see it on the whole screen, as the `view` command shows a file. A link with a picture in it says `(i: its picture)` on the status line: `i` shows the picture, Enter still follows the link. The right button's menu has **View image** for both. Any key or click goes back to the page.

PNG, JPEG and BMP pictures can be shown (scaled to fit, dithered to the screen's 256 colours); GIF, WebP and SVG can't, and say so. The picture is loaded into PSRAM and written to `/WWW/IMAGE.TMP` on the drive for a moment while it is decoded. Pictures without alt text (often spacers and icons) aren't shown on the page.

Bookmarks, saving and downloading

- **a** adds the page to `/WWW/BOOKMARK.HTM` (an HTML page of links: edit it with `edit` if you like); **v** opens it.
- **d** downloads the chosen link to the drive (it asks for the name). Following a link to something that isn't a page (a ZIP, a picture, a program) asks the same. Files up to 4 MB go through PSRAM; for bigger ones use `wget`.
- **s** saves the page's HTML, **p** the page as text (CRLF lines, as on the screen).
- `/WWW/HOME.HTM` on the drive replaces the start page: any HTML page, for example your bookmarks copied there.



What it does and doesn't do

- HTML is read in one pass: paragraphs, headings, lists (numbered and nested), definition lists, quotes, `<pre>` text, rules, tables (each row on a line, the cells side by side), links, anchors (`#name`), frames and meta refreshes (as links), forms, entities and UTF-8 or Latin-1 text. Characters the screen's font (code page 437) lacks are shown as the nearest one (accents dropped, curly quotes straight) or `?`.
- Not done: pictures in the page (they are shown on their own, see Pictures), JavaScript, CSS, layout tables side by side, text areas of more than one line, file uploads, compressed replies (the browser doesn't ask for them), HTTP/2.
- Pages up to 4 MB are loaded; bigger ones are cut there.
- https: the certificate is checked against `CACERT.PEM` (see the *Network Tools* manual). If it can't be checked, the browser asks whether to load the page anyway (Yes, No, or Always for this session); View > Check certificates turns checking off and on.
- It identifies itself as lynx, so sites send their simple pages.

Memory

Everything the browser keeps is in PSRAM, and only while it is open: the pages it has shown (up to 3 MB of them, so Back doesn't load them again; older ones are loaded again when needed), each page laid out, the history (40 pages), the cookies. While a page loads, the network uses the same SRAM as `wget` (a 16 KB receive buffer, and about 40 KB more for https).

Testing on a PC

`tools/wwwtest/uitest.sh /absolute/out/folder` builds the browser for a PC with its real screen, menus, dialogs and fetcher against a pretend web server, drives it with keys and the mouse (pages, links, the right button's menu, pictures, a redirect that sets a cookie, the forms by GET and POST, find, the source, a download, bookmarks, the history, help), checks what the screen shows, and writes the screenshots in this manual (`www_*.png`). It also lays out thousands of random and cut-off pages and checks URLs are resolved as browsers do, all under AddressSanitizer.